

Лабораторная работа № 11

Основы администрирования операционных систем

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Получить навыки работы с загрузчиком системы GRUB2.

1. Продемонстрировать навыки по изменению параметров GRUB и записи изменений в файл конфигурации
2. Продемонстрировать навыки устранения неполадок при работе с GRUB
3. Продемонстрировать навыки работы с GRUB без использования root

Выполнение работы

Получение прав администратора, установка параметра

```
GRUB_TIMEOUT=10
GRUB_DISTRIBUTOR="$(sed 's, release .*$,,g' /etc/system-release)"
GRUB_DEFAULT=saved
GRUB_DISABLE_SUBMENU=true
GRUB_TERMINAL_OUTPUT="console"
GRUB_CMDLINE_LINUX="resume=/dev/mapper/rl_svivanov1-swap rd.lvm.lv=rl_svivanov1/
root rd.lvm.lv=rl_svivanov1/swap rhgb quiet"
GRUB_DISABLE_RECOVERY="true"
GRUB_ENABLE_BLSCFG=true
~
~
```

Рис. 1: Получение прав администратора, установка параметра отображения меню загрузки

Запись изменений в GRUB2

```
[root@svivanov1 ~]# grub2-mkconfig > /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
[root@svivanov1 ~]# █
```

Рис. 2: Запись изменений в GRUB2

Перезагрузка и просмотр загрузочных сообщений

```
[ OK ] Finished Load Kernel Modules.  
[ OK ] Finished Generate network units from Kernel command line.  
[ OK ] Finished Remount Root and Kernel File Systems.  
[ OK ] Reached target Preparation for Network.  
[ OK ] Reached target Swaps.  
        Mounting FUSE Control File System...  
        Mounting Kernel Configuration File System...  
        Starting Flush Journal to Persistent Storage...  
        Starting Load/Save OS Random Seed...  
        Starting Apply Kernel Variables...  
        Starting Create Static Device Nodes in /dev...  
[ 3.300198] systemd-journald[629]: Received client request to flush runtime journal.  
[ OK ] Finished Coldplug All udev Devices.  
[ OK ] Mounted FUSE Control File System.  
[ OK ] Mounted Kernel Configuration File System.  
[ OK ] Finished Load/Save OS Random Seed.  
        Starting Wait for udev To Complete Device Initialization...  
[ OK ] Finished Apply Kernel Variables.
```

Рис. 3: Перезагрузка и просмотр загрузочных сообщений

Редактирование в меню GRUB

```
GRUB version 2.06

load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-5.14.0-427.42.1.el9_4.x86_64 root=/dev/mapper/rl_svivanov1-root ro resume=/dev/mapper/rl_svivanov1-swap rd.lvm.lv=rl_svivanov1/r\
oot rd.lvm.lv=rl_svivanov1/swap systemd.unit=rescue.target crashkernel=1G-4\
G:192M,4G-64G:256M,64G-:512M
initrd ($root)/initramfs-5.14.0-427.42.1.el9_4.x86_64.img $tuned_initrd

Minimum Emacs-like screen editing is supported. TAB lists
completions. Press Ctrl-x or F10 to boot, Ctrl-c or F2 for
a command-line or ESC to discard edits and return to the GRUB menu.
```

Рис. 4: Редактирование параметров загрузки в меню GRUB

Просмотр списка всех файлов модулей

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
proc-sys-fs-binfmt_misc.automount	loaded	active	waiting	Arbitrary Executable File Formats File System
sys-devices-pci0000:00-0000:00:01.1-ata3-host2-target2:0:0-2:0:0:0-block-sr0.device	loaded	active	plugged	UBOX_CD-ROM UBox_Gas_7.0.14
sys-devices-pci0000:00-0000:00:03.0-net-emp8s3.device	loaded	active	plugged	82548EM Gigabit Ethernet Controller (PRO-
sys-devices-pci0000:00-0000:00:05.0-sound-card0-control00.device	loaded	active	plugged	/sys/devices/pci0000:00/0000:00:05.0/sound
sys-devices-pci0000:00-0000:00:04.0-ata1-host0-target0:0:0-0:0:0:0-block-sda1.device	loaded	active	plugged	UBOX_HARDDISK 1
sys-devices-pci0000:00-0000:00:04.0-ata1-host0-target0:0:0-0:0:0:0-block-sda2.device	loaded	active	plugged	UBOX_HARDDISK 2
sys-devices-pci0000:00-0000:00:04.0-ata1-host0-target0:0:0-0:0:0:0-block-sda.device	loaded	active	plugged	UBOX_HARDDISK
sys-devices-platform-serial18250-tty-ttyS0.device	loaded	active	plugged	/sys/devices/platform/serial18250/tty/ttyS
sys-devices-platform-serial18250-tty-ttyS1.device	loaded	active	plugged	/sys/devices/platform/serial18250/tty/ttyS
sys-devices-platform-serial18250-tty-ttyS2.device	loaded	active	plugged	/sys/devices/platform/serial18250/tty/ttyS
sys-devices-platform-serial18250-tty-ttyS3.device	loaded	active	plugged	/sys/devices/platform/serial18250/tty/ttyS
sys-devices-virtual-block-dm0-dm240.device	loaded	active	plugged	/sys/devices/virtual/block/dm-0
sys-devices-virtual-block-dm0-dm241.device	loaded	active	plugged	/sys/devices/virtual/block/dm-1
sys-module-configfs.device	loaded	active	plugged	/sys/module/configfs
sys-module-fuse.device	loaded	active	plugged	/sys/module/fuse
sys-subsystem-net-devices-emp8s3.device	loaded	active	plugged	82548EM Gigabit Ethernet Controller (PRO-
-mount	loaded	active	mounted	Root Mount
boot.mount	loaded	active	mounted	/boot
dev-hugepages.mount	loaded	active	mounted	Huge Pages File System
dev-mqueue.mount	loaded	active	mounted	POSIX Message Queue File System
run-credentials-system\x2dsysctl.service.mount	loaded	active	mounted	/run/credentials/systemd-sysctl.service
run-credentials-system\x2dtmpfiles\x2dsetup.service.mount	loaded	active	mounted	/run/credentials/systemd-tmpfiles-setup.s
run-credentials-system\x2dtmpfiles\x2dsetup\x2ddev.service.mount	loaded	active	mounted	/run/credentials/systemd-tmpfiles-setup-d
sys-fs-fuse-connections.mount	loaded	active	mounted	FUSE Control File System
sys-kernel-config.mount	loaded	active	mounted	Kernel Configuration File System
sys-kernel-debug.mount	loaded	active	mounted	Kernel Debug File System
sys-kernel-tracing.mount	loaded	active	mounted	Kernel Trace File System
systemd-ask-password-plymouth.path	loaded	active	waiting	Forward Password Requests to Plymouth Dir
init.scope	loaded	active	running	System and Service Manager
alsa-state.service	loaded	active	running	Manage Sound Card State (restore and stor
dracut-shutdown.service	loaded	active	exited	Restore /run/inittmpfs on shutdown
tmpd-static-nodes.service	loaded	active	exited	Create List of Static Device Nodes

Рис. 5: Просмотр списка всех файлов модулей, которые загружены в настоящее время

Задействованные переменные среды оболочки

```
[root@svivanov1 ~]# systemctl show-environment  
LANG=en_US.UTF-8  
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin  
[root@svivanov1 ~]#
```

Рис. 6: Просмотр задействованных переменных среды оболочки

Редактирование в меню GRUB

```
load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-5.14.0-427.42.1.el9_4.x86_64 root=/dev/mapper/rl_svivanov1-root ro resume=/dev/mapper/rl_svivanov1-swap rd.lvm.lv=rl_svivanov1/rd.lvm.lv=rl_svivanov1/swap systemd.unit=emergency.target crashkernel=1G-4G:192M,4G-64G:256M,64G-:512M
initrd ($root)/initramfs-5.14.0-427.42.1.el9_4.x86_64.img $tuned_initrd
```

Рис. 7: Редактирование параметров загрузки в меню GRUB

Просмотр списка всех загруженных файлов модулей

UNIT	LOAD	ACTIVE	SUB	DESCRIPTION
dev-cdrom.device	loaded	activating	tentative	/dev/cdrom
dev-disk-by\x2ddiskseq-1.device	loaded	activating	tentative	/dev/disk/by-diskseq/1
dev-disk-by\x2ddiskseq-3.device	loaded	activating	tentative	/dev/disk/by-diskseq/3
dev-disk-by\x2did-ata\x2dUBOX_CD-80	loaded	activating	tentative	/dev/disk/by-id/ata-UBOX_CD-80
dev-disk-by\x2did-ata\x2dUBOX_HARDDISK_0b595ab67\x2d494a9dc65.device	loaded	activating	tentative	/dev/disk/by-id/ata-UBOX_HARDDISK_0b595ab67\x2d494a9dc65
dev-disk-by\x2did-ata\x2dUBOX_HARDDISK_0b595ab67\x2d494a9dc65\x2dpart1.device	loaded	activating	tentative	/dev/disk/by-id/ata-UBOX_HARDDISK_0b595ab67\x2d494a9dc65-part1
dev-disk-by\x2did-ata\x2dUBOX_HARDDISK_0b595ab67\x2d494a9dc65\x2dpart2.device	loaded	activating	tentative	/dev/disk/by-id/ata-UBOX_HARDDISK_0b595ab67\x2d494a9dc65-part2
dev-disk-by\x2dlabel-UBOX_GNs_7_0.14.device	loaded	activating	tentative	/dev/disk/by-label/UBOX_GNs_7_0.14
dev-disk-by\x2dpartuuid-8c000d44e\x2d001.device	loaded	activating	tentative	/dev/disk/by-partuuid/8c000d44e0000001
dev-disk-by\x2dpartuuid-8c000d44e\x2d002.device	loaded	activating	tentative	/dev/disk/by-partuuid/8c000d44e0000002
dev-disk-by\x2dpath-pci\x2d0000:00:01.1\x2data\x2d2.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:01.1-ata-2
dev-disk-by\x2dpath-pci\x2d0000:00:01.1\x2data\x2d2.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:01.1-ata-2
dev-disk-by\x2dpath-pci\x2d0000:00:0d.0\x2data\x2d1.0.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:0d.0-ata-1.0
dev-disk-by\x2dpath-pci\x2d0000:00:0d.0\x2data\x2d1.0\x2dpart1.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:0d.0-ata-1.0-part1
dev-disk-by\x2dpath-pci\x2d0000:00:0d.0\x2data\x2d1.0\x2dpart2.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:0d.0-ata-1.0-part2
dev-disk-by\x2dpath-pci\x2d0000:00:0d.0\x2data\x2d1.0.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:0d.0-ata-1.0
dev-disk-by\x2dpath-pci\x2d0000:00:0d.0\x2data\x2d1\x2dpart1.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:0d.0-ata-1.0-part1
dev-disk-by\x2dpath-pci\x2d0000:00:0d.0\x2data\x2d1\x2dpart2.device	loaded	activating	tentative	/dev/disk/by-path/pci-0000:00:0d.0-ata-1.0-part2
dev-disk-by\x2duuid-2024\x2d01\x2d15\x2d14\x2d40\x2d13\x2d493.device	loaded	activating	tentative	/dev/disk/by-uuid/2024-01-15-144013493
dev-disk-by\x2duuid-4075baae\x2d2582\x2d244dc3\x2d404d6\x2d734ae863fd.device	loaded	activating	tentative	/dev/disk/by-uuid/4075baae-2582-244dc3-404d6-734ae863fd
dev-sda.device	loaded	activating	tentative	/dev/sda
dev-sda1.device	loaded	activating	tentative	/dev/sda1
dev-sda2.device	loaded	activating	tentative	/dev/sda2
dev-sr0.device	loaded	activating	tentative	/dev/sr0
dev-ttyS0.device	loaded	activating	tentative	/dev/ttyS0
dev-ttyS1.device	loaded	activating	tentative	/dev/ttyS1
dev-ttyS2.device	loaded	activating	tentative	/dev/ttyS2
dev-ttyS3.device	loaded	activating	tentative	/dev/ttyS3
sys-devices-pci0000:00:0000:00:01.1-ata2-target1:0:0-1:0:0:0-block-sr0.device	loaded	activating	tentative	/sys/devices/pci0000:00:0000:00:01.1-ata2-target1:0:0-1:0:0:0-block-sr0
sys-devices-pci0000:00:0000:00:03.0-net-emp0s3.device	loaded	activating	tentative	/sys/devices/pci0000:00:0000:00:03.0-net-emp0s3
sys-devices-pci0000:00:0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda-sda1.device	loaded	activating	tentative	/sys/devices/pci0000:00:0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda-sda1
sys-devices-pci0000:00:0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda-sda2.device	loaded	activating	tentative	/sys/devices/pci0000:00:0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda-sda2
sys-devices-pci0000:00:0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda.device	loaded	activating	tentative	/sys/devices/pci0000:00:0000:00:0d.0-ata3-host2-target2:0:0-2:0:0:0-block-sda
sys-devices-platform-serial18250-tty-ttyS0.device	loaded	activating	tentative	/sys/devices/platform/serial18250-tty-ttyS0
sys-devices-platform-serial18250-tty-ttyS1.device	loaded	activating	tentative	/sys/devices/platform/serial18250-tty-ttyS1
sys-devices-platform-serial18250-tty-ttyS2.device	loaded	activating	tentative	/sys/devices/platform/serial18250-tty-ttyS2
sys-devices-platform-serial18250-tty-ttyS3.device	loaded	activating	tentative	/sys/devices/platform/serial18250-tty-ttyS3
sys-devices-virtual-block-dm\x2d0.device	loaded	active	plugged	/sys/devices/virtual/block/dm-0

Рис. 8: Просмотр списка всех загруженных файлов модулей

Редактирование в меню GRUB

```
GRUB version 2.06

load_video
set gfxpayload=keep
insmod gzio
linux ($root)/vmlinuz-5.14.0-427.42.1.el9_4.x86_64 root=/dev/mapper/rl_svivanov1-root ro resume=/dev/mapper/rl_svivanov1-swap rd.lvm.lv=rl_svivanov1/r\
oot rd.lvm.lv=rl_svivanov1/swap rd.break_crashkernel=1G-4G:192M,4G-64G:256M\
,64G-:512M
initrd ($root)/initramfs-5.14.0-427.42.1.el9_4.x86_64.img $tuned_initrd
```

Рис. 9: Редактирование параметров загрузки в меню GRUB

```
switch_root:/# mount -o remount,rw /sysroot  
switch_root:/# chroot /sysroot  
sh-5.1#
```

Рис. 10: Получение доступа к системному образу и новый корневой каталог

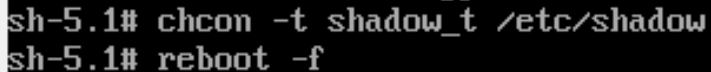
```
sh-5.1# passwd
Changing password for user root.
New password:
Retype new password:
passwd: all authentication tokens updated successfully.
sh-5.1#
```

Рис. 11: Установка нового пароля

Загрузка политики SELinux

```
sh-5.1# load_policy -i
[ 214.224175] audit: type=1404 audit(1731664736.243:2): enforcing=1 old_enforcing=0 auid=4294967295 ses=4294967295
[ 214.319736] SELinux: policy capability network_peer_controls=1
[ 214.320110] SELinux: policy capability open_perms=1
[ 214.320450] SELinux: policy capability extended_socket_class=1
[ 214.320784] SELinux: policy capability always_check_network=0
[ 214.321111] SELinux: policy capability cgroup_seclabel=1
[ 214.321448] SELinux: policy capability nnp_nosuid_transition=1
[ 214.321775] SELinux: policy capability genfs_seclabel_symlinks=1
[ 214.654691] audit: type=1403 audit(1731664736.673:3): auid=4294967295 ses=4294967295 lsm=selinux res=1
sh-5.1#
```

Рис. 12: Загрузка политики SELinux



```
sh-5.1# chcon -t shadow_t /etc/shadow
sh-5.1# reboot -f
```

Рис. 13: Установка типа контекста и перезагрузка

Вывод

В ходе выполнения лабораторной работы были получены навыки работы с загрузчиком системы GRUB2.

<https://esystem.rudn.ru/mod/page/view.php?id=1098933>